

$$\forall \varepsilon \in \mathbb{R}^*_+ \; \exists \eta > 0 \; |x-x_0| \leq \eta \Longrightarrow |f(x)-f(x_0)| \leq \varepsilon$$

$$\det \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ a_{n1} & \cdots & \cdots & a_{nn} \end{bmatrix} \stackrel{\text{def}}{=} \sum_{\sigma \in \mathcal{S}_n} \varepsilon(\sigma) \prod_{k=1}^n a_{k\sigma(k)}$$

$${}^{\beta}_{\alpha}\begin{pmatrix}a&b\\c&d\end{pmatrix}_{\gamma}^{\delta}$$

$$\int_0^\infty x^{2n}e^{-ax^2}\,dx=\frac{2n-1}{2a}\int_0^\infty x^{2(n-1)}e^{-ax^2}\,dx=\frac{(2n-1)!!}{2^{n+1}}\sqrt{\frac{\pi}{a^{2n+1}}}$$

$$\int_a^b f(x)\,dx=(b-a)\sum_{n=1}^\infty\sum_{m=1}^{2^n-1}(-1)^{m+1}2^{-n}f(a+m\,(b-a)\,2^{-n})$$

$$\int_{-\pi}^{\pi} \sin(\alpha x) \sin^n(\beta x) dx = \left\{ \begin{array}{ll} (-1)^{(n+1)/2} (-1)^{m \frac{2\pi}{2^n}} \binom{n}{m} & n \text{ odd, } \alpha = \beta (2m-n) \\ 0 & \text{otherwise} \end{array} \right.$$

$$L=\int_a^b\sqrt{\left|\sum_{i,j=1}^ng_{ij}(\gamma(t))\left(\frac{d}{dt}x^i\circ\gamma(t)\right)\left(\frac{d}{dt}x^j\circ\gamma(t)\right)\right|}dt$$

$$\begin{aligned}s&=\int_a^b\left\|\frac{d}{dt}\vec{r}(u(t),v(t))\right\|\,dt\\&=\int_a^b\sqrt{u'(t)^2\,\vec{r}_u\cdot\vec{r}_u+2u'(t)v'(t)\,\vec{r}_u\cdot\vec{r}_v+v'(t)^2\,\vec{r}_v\cdot\vec{r}_v}\,dt.\end{aligned}$$